

Week of August 16

Project: iGEM KU Leuven 2023

Author: Jane (Junfei) Chen

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MERCREDI 16/08/2023

- **plasmid extraction:**

Miniprep using NucleoSpin kit of pBevy-Nat plasmid transformed into E coli Top10 cells

 PT4022-1.pdf

result: pBEVY-Nat was extracted for transformation and final plasmid concentrations were: 500 ng/uL and 528 ng/uL

- **E.coli stocks preparation:**

result: Made glycerol stocks of E coli cells (20% glycerol) and stored at - 80°C.

participants: Shekoufeh M M Sadeghi, Vlad Batagui and Junfei Chen.

JEUDI 17/08/2023

- **Yarrowia culture:**

Yarrowia stocks were received as pieces of agar containing the yeast and cultured on YPD plates for future experiments. One agar piece was dissolved into YPD and a YPD-agar plate was streaked to generate stock. An overnight liquid culture was also generated.

Protocol: Yarrowia was picked from the stock and plated using the streak plate method and incubated overnight at 30°C.

Result: Yarrowia culture was ready for following steps.

participant: Vlad Batagui

VENDREDI 06/10/2023

25 % glycerol stocks were made for the Yarrowia lipolytica strain.

Week of August 21

Project: iGEM KU Leuven 2023

Author: Vlad Batagui

Entry Created On: 21 Aug 2023 08:55:23 UTC

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LUNDI 21/08/2023

- **Yarrowia inoculation:**

the overnight culture of Yarrowia was inoculated for use in growth curve measurements.

protocol:

1. one single colony was picked from the cultured plate and inoculated in 3 ml of YPD medium.
2. the tube was incubated at 30°C overnight.

participants: vlad Batagui

MARDI 22/08/2023

- **Growth curve measurements:**

1. OD600 of overnight inoculation of Yarrowia was measured= 6.89 And was diluted to 0.1.
2. Media were prepared in erlenmyer flasks:

growth on glucose: 50 ml YPD (2% glucose)

growth on oil: 48.15 ml YPD + 1.85 ml oil. (3.7% oil)

3. the flasks were placed in incubator at 200 rpm at 30°C.

4. OD of both cultures was measured every hour at the beginning and shifted to every 1.5 hours after observing the growth of our strain. the measurements continued for 10 hours measuring a serial dilution of 1:10 and 1:100. this serial dilution was made to generate additional data points for the growth curve and to account as well for the errors of the machine.

result: the experiment was cut short as the medium used for oil growth contained glucose as well. and the protocol was still not well defined as the Yarrowia growth and methods used to measure the growth differed between strains and articles.

- **Yarrowia inoculation.**

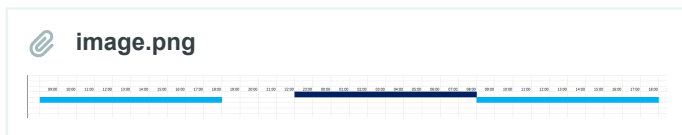
participant: Shekoufeh M M Sadeghi

MERCREDI 23/08/2023

- **growth curve measurements:**

1. the medium preparation was followed as yesterday substituting YPD for YP for the oil medium.
2. OD of inoculated overnight was measured: 6.4 A => to dilute to 0.1 A we added 0.780 ml into the cultures.
3. cultures were put into the incubator at 9:20 a.m. and at 200 rpm and 30°C.
4. ODs were measured every 2 hours from 9 to 19 for 10 hours and the results were recorded.
5. each sample was prepared with a serial dilution of 1: 10 and 1: 100 for OD measurements by mixing 100 ul of culture and 900ul of medium (YP was used for oil and YPD for Glu).
6. at 19:00 the last data point was taken and the overnight cultures were prepared. the seed culture's OD was at 11.795 and was diluted to 0.1.

the overnight cultures were used to cover the data lost during night time so that the two cultures can together cover the curve so that there is minimal data loss and the data matched and the attempt was successful.



the picture represents how the overnight and day cultures cover our growth curve.

result: the overall strategy of the growth curve measurements was drawn¹ and the curve was started.

participant: Shekoufeh M M Sadeghi

JEUDI 24/08/2023

- **growth curve measurements:**

the same process was followed and OD measurements were performed every 2 hours for 10 hours on all 4 cultures.

as the Yarrowia growth was still going the measurements continued for an additional day.

participants: Shekoufeh M M Sadeghi and Vlad Batagui.

VENDREDI 25/08/2023

- **growth curve measurements:**

the same process continued for a 3rd day. the curve was stopped here as both cultures had reached the stationary phase.

result: Yarrowia growth was better on oil compared to glucose and the reasoning was that based on the moles of carbon, 2% glucose is not equivalent to 3.7% of oil and to get an equivalent the concentration of glucose needs to be at 5% ² which was remedied later on in the experiments.

participants: Shekoufeh M M Sadeghi and Vlad Batagui.

1. Du HX, Xiao WH, Wang Y, Zhou X, Zhang Y, Liu D, Yuan YJ. Engineering Yarrowia lipolytica for Campesterol Overproduction. PLoS One. 2016 Jan 11;11(1):e0146773. doi: 10.1371/journal.pone.0146773. PMID: 26751680; PMCID: PMC4709189.
2. Ryan Sestric, Garret Munch, Nazim Cicek, Richard Sparling, David B. Levin, Growth and neutral lipid synthesis by Yarrowia lipolytica on various carbon substrates under nutrient-sufficient and nutrient-limited conditions, Bioresource Technology, Volume 164, 2014, Pages 41-46, ISSN 0960-8524

Week of Sep 11

Project: iGEM KU Leuven 2023

Author: Jane (Junfei) Chen

Entry Created On: 13 Sep 2023 07:15:39 UTC

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LUNDI 11/09/2023

- **yarrowia inoculation:**

one single colony from Yarrowia plate was inoculated in 3ml of YPD and cultured overnight at 30°C.

result: Yarrowia was inoculated and ready for use for growth curve measurements.

participant: Shekoufeh M M Sadeghi

Yarrowia plate was made on 10/09/2023 Sunday by Vlad Batagui.

MARDI 12/09/2023

- **growth curve measurement:**

OD was measured every 2 hours starting from 9 a.m. to 7 p.m.

overnight cultures were prepared at 7 p.m.

result: the new growth curve measurements were done to create replicates for the initial growth curve to improve the accuracy of the curve. this was requested from the dry lab team who were creating the growth curve. however the new curve was generated with the same variables but with 250 rpm at 30°C.

participants: Shekoufeh M M Sadeghi, Junfei Chen

MERCREDI 13/09/2023

- **growth curve measurement**

OD was measured every 2 hours starting from 9 a.m. to 7 p.m.

participants: Shekoufeh M M Sadeghi, Junfei Chen

- **S. cereavisae gDNA extraction:**

protocol:

 **SOP_ Isolation of gDNA and ethanol precipitation.docx**

result: the S cerevisiae gDNA was prepared for transformation.

participants: Junfei Chen and Shekoufeh M M Sadeghi.

- **PCR amplification of hydrophobin sequences and plasmid:**

protocol:

 Q5 PCR protocol NEB.pdf

Table1		
	A	B
1	pBEVY Fwd	CACCATCACCATTAATACCGAGCTCGAATTCGACAC
2	pBEVY Rev	GCAGTAAAAATTGAAGGAAATCTCATCCCGGGGAGTTGATTGTATGC
3	MF Fwd	GCATACAATCAACTCCCCGGGATGAGATTTCTTCAATTTTTACTGCAG
4	MF Rev	TGAACCTCCAGAACCGCCACTTCTTTTATCCAAAGATACCCCTTCTTC
5	HFBI Fwd	AGTGGCGGTTCTGGAGGTTTCATCTAACGGCAATGGTAACGTTTGtcc
6	HFBII Rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGCGccccaaccgctgtttg
7	HFBII Fwd	AGTGGCGGTTCTGGAGGTTCAAAGTTCTTCACTGCAGCAGCCC
8	HFBII Rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGAGCAGCACCAACAGCAGC
9	MBSP1 Fwd	AGTGGCGGTTCTGGAGGTTTCATCGGACCAATACCTTGACTTTG
10	MBSP1 Rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGCGTGGAATCCTGGCCC

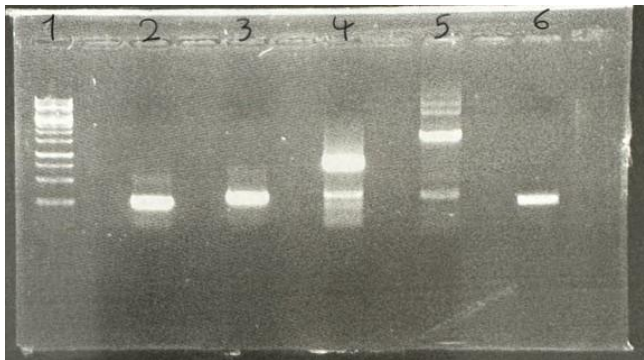
PCR for hydrophobins HFBI, HFBII, MF and MBSP1 and pBEVY was performed using Q5.
conditions:
geneblocks: denature: 98°C for 10 sec, annealing: 55°C for 30 sec, elongation: 72°C for 30 sec
pBEVY backbone: same but with elongation of 2 min and 20 sec.
dilutions: primers are diluted to 10ng/ul, same as template.

result: the genes and plasmid are amplified and ready for vector construction.
participant: Junfei Chen

JEUDI 14/09/2023

- growth curve measurement
- OD was measured every 2 hours starting from 9 a.m. to 7 p.m.
participants: Shekoufeh M M Sadeghi, Junfei Chen

- Gel electrophoresis for PCR product:

gel Q5 amplification.jpeg

1= ladder, 2=HFB1, 3= HFB2, 4= MBSP1, 5=pBEVY, 6= MF

result: all amplicons were at the correct length however the plasmid was not the expected length of 7 kbp but lower and PCR had to be reperformed.

participant: Junfei Chen

- **reperforming the Q5 PCR**

Theoretical annealing temp of 68°C and suggested annealing temp of 55°C, elongation temp at 72°C for 2min and 20 sec.

participant: Junfei Chen and Shekoufeh M M Sadeghi

- **gel electrophoresis:**

longer, wider gel with only 2 uL of DNA was loaded.

result: pBEVY still shows the same issue therefore our advisor suggested redoing PCR with 7min30sec elongation time and using 0.7% agarose gel with smaller size.

participant: Junfei Chen and Shekoufeh M M Sadeghi

VENDREDI 15/09/2023

- **performing Q5 PCR:**

PCR was redone for pBEVY with 72°C and 7 min and 30 sec elongation time.

participant: Shekoufeh M M Sadeghi

Week of sep 18

Project: iGEM KU Leuven 2023

Author: Shokoofeh Mms

Entry Created On: 18 Sep 2023 16:20:14 UTC

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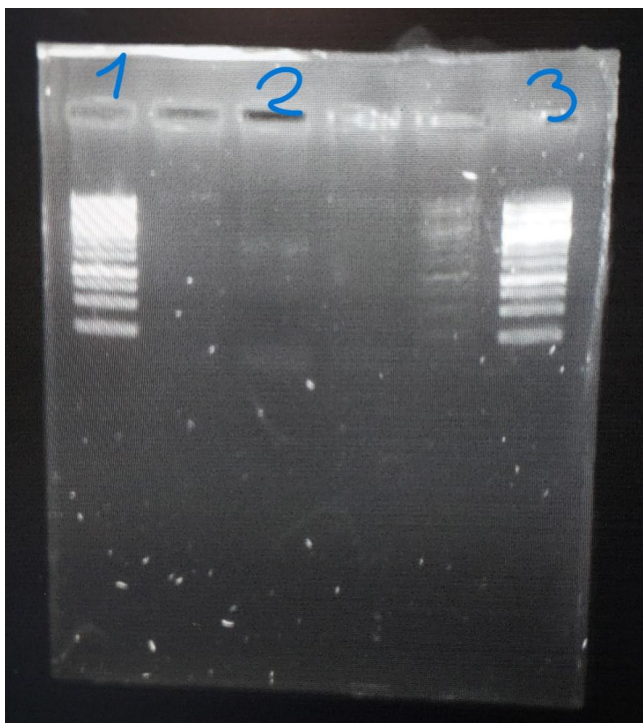
Export Generated On: 10 Oct 2023 18:13:44 UTC

LUNDI 18/09/2023

- **gel electrophoresis:**

prepared 0.89% gel and ran pBEVY amplicon.

 Q5 pBEVY amplification



1 & 3= DNA ladders, 2= PCR pBEVY amplicon

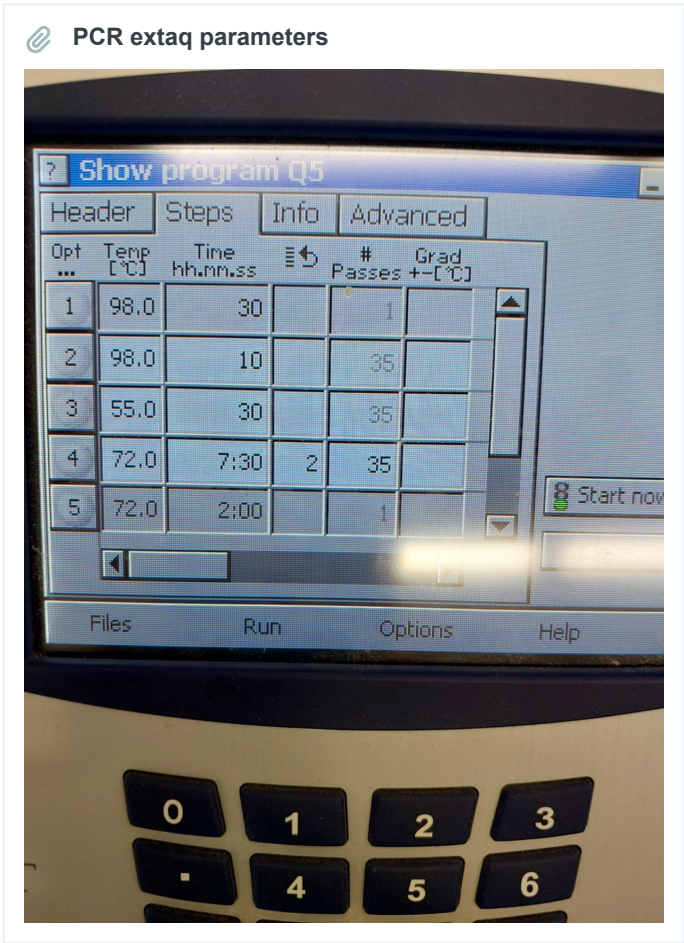
result: higher band comes to around 2kb and is therefore still not at 7kb and also very faint.

participant: Shekoufeh M M Sadeghi

- **pBEVY PCR amplification:**

Extaq enzyme was used instead of Q5.

PCR parameters were unchanged:



protocol:

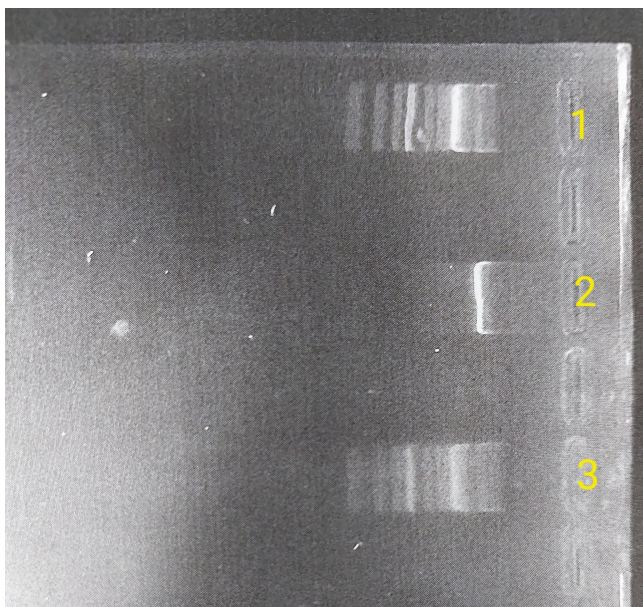


participant: Shekoufeh M M Sadeghi

MARDI 19/09/2023

- gel electrophoresis:
ran the pBEVY amplicon on 0.89% agarose gel for 10 minutes.

ExTaq pBEVY amplification



1 & 3= DNA ladder (1kb), 2= pBEVY.

result: The pBEVY comes out correctly with Extaq.

note: getting the confirmation from Extaq that there is in fact nothing wrong with the primers; proceeded to rerun the PCR with Q5 however using a gradient PCR to test different annealing temperatures.

participant: Shekoufeh M M Sadeghi

MERCREDI 20/09/2023

- gradient PCR:

ran PCR with gradient annealing temperature ranging from 54 to 68 C.

- gel electrophoresis

Q5 gradient PCR pBEVY.jpeg



1= ladder, 2=56.6°C, 3=58.8°C, 4=61.2°C, 5=65.6°C, 6=67.4°C

result: we did not see the band with the desired length with any of the annealing temperatures, therefore, we proceeded to use the pBEVY amplicon amplified using Extaq for vector construction.

participant: Junfei Chen

JEUDI 21/09/2023

- plasmid digestion:

On this page: Entry

the pBEVY amplified by extaq was digested using dpn1 and rcutsmart buffer from NEB at 37°C for 2 hours.

participant: Shekoufeh M M Sadeghi

VENDREDI 22/09/2023

- **DNA purification of plasmid and amplicons:**

the digested pBEVY, and MF, HFB1, HFB2, MBSP1 amplicons were taken for DNA purification to prepare for vector construction.

protocol:

 TFS-Assets_LSG_manuals_MAN0012662_GeneJET_PCR_Purification_UG.pdf

- **vector assemblies:**

after purification assembly was performed for each of the biosurfactants: pBEVY+MF+biosurfactant at 50 deg at 800 rpm for 1 hour.

protocol:

- I. 5uL of NEB buffer, 1uL of cut vector, 2uL of HFBII , 2uL of MF were added in an eppendorf tube (same reaction was prepared for HFBI and MBSP1).
- II. the solution was incubated at 50°C for 1 hour

result: vectors were assembled and were ready for transformation.

- **heat shock transformation:**

transformation of competent ecoli cells from the stock of the lab was performed for each of our 3 vectors.

protocol:

1. thaw Ecoli competent cell on ice
2. add 4uL of the assembly and vortex into the cell
3. incubate 20 min on ice
4. heat shock at 42°C for 45 sec
5. add 1mL of LB medium and incubate 1 hour at 37°C with 800 rpm
6. plate cells on LB+amp medium using glass beads to get single colonies.

3 plates for each transformed batch was cultured:

1. 10ul cell/ 90 ul sterile water
2. 100 ul cells
3. rest of cells (were spun down 4000 rpm, 2 min, most of the supernatant was taken out and the cells were resuspended and plated).

the plates were put in the 37 deg incubator around 13 p.m. and were taken out at 9 a.m.the next day.

result: E.coli cells were now transformed with our vectors containing our biosurfactant genes.

participants: Junfei Chen and Shekoufeh M M Sadeghi

Week of Sep 25

Project: iGEM KU Leuven 2023

Author: Jane (Junfei) Chen

Entry Created On: 25 Sep 2023 14:39:34 UTC

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DIMANCHE 24/09/2023

- **Yarrowia inoculation:**

1 colony was picked from yarrowia culture and was inoculated in 3ml YPD for growth curve measurements 20:00 p.m.
participant: Junfei Chen

LUNDI 25/09/2023

- **growth curve measurement:**

growth curve measurements had to be repeated as the carbon mols for our glucose concentration were not equivalent to the oil concentration as explained previously.

new medium was prepared with 5% glucose for the growth on glucose and 3.7% oil for the oil culture:

- 5% glucose YP: 43.75 ml of YP + 6.25 ml 40% glucose
- 3% oil YP: 1.85ml oil + 48.15 YP

OD of the inoculated Yarrowia was measured and diluted to 0.1. the cultures were prepared in erlenmyer flasks and placed in the incubator at 30°C at 200 rpm.

2 replicates of each medium were prepared for additional data.

note: because the growth of oil was still much faster than growth on glucose there was a possibility that the oil was influencing the OD measurements: therefore the data was checked at 17:00 p.m. by centrifuging 1 ml of the first replicate of oil and glucose to compare the amount of cells present in each sample. by analysing the pallet we saw that the amount of cells in oil medium were in fact much greater compared to glucose (seemed around 5 times more which corresponds with the OD measurements so the reason for higher OD is not because of the oil but cells).

growth curve measurements were performed every 2 hours from 9 a.m. to 19 p.m.

night cultures were prepared following the same protocol as with day cultures and were placed in the incubator.

participants: Junfei Chen, Shekoufeh M M Sadeghi and JeongHo Choi

MARDI 26/09/2023

- **growth curve measurements:**

growth measurements were performed every 2 hours from 9 a.m. to 19 p.m.

participants: Junfei Chen, Shekoufeh M M Sadeghi and JeongHo Choi

- **selecting primers for colony PCR:**

primers for colony PCR were confirmed:

D-826= forward: 5' GTTGTTGTCTCACCATATCC 3'

D-327= reverse: 5' CTGTTTGTGGAGGATGCCGTA 3'

MERCREDI 27/09/2023

- **growth measurements**

result: growth curve data was collected and was stopped after 3 days (60 hours) such as our previous experiment had demonstrated.

participants: Junfei Chen, Shekoufeh M M Sadeghi and JeongHo Choi

result:

 growth curve data.xlsx

- **colony PCR on transformed E.coli cells**

PCR was performed using taq enzyme on 8 colonies from each hydrophobin transformation (total 24 samples).

protocol:

from each only the plate with the least amount of colonies was selected for transformation:

from HFB1 and HFB2 only the 10ul sample: 90ul water plate was used but for MBSP1 100ul sample plate was also used since there were not enough colonies on the 10ul sample: 90ul water.

colonies were picked from the plate with a 200ul pipet tip and first transferred on a new plate (LB+amp) and afterward placed in reaction tubes.

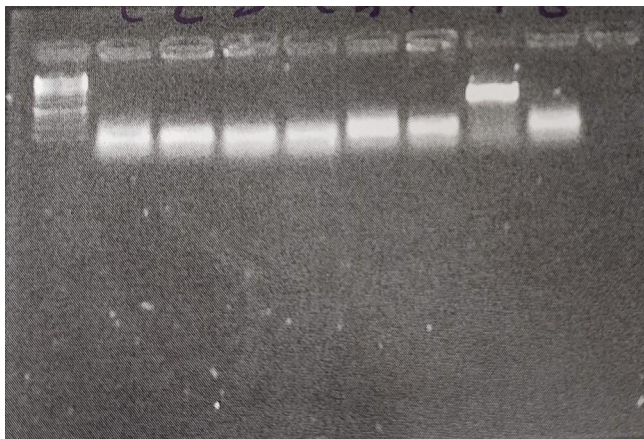
 extaq PCR.pdf

participants: Junfei Chen and Shekoufeh M M Sadeghi

- **gel electrophoresis**

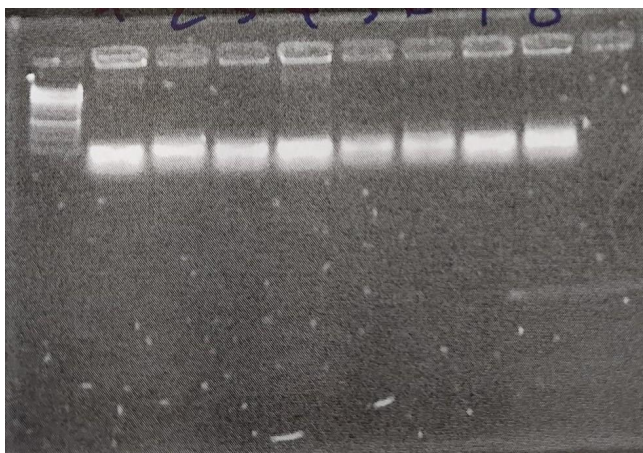
gel electrophoresis was performed to visualize the data.

 HFBII.jpeg



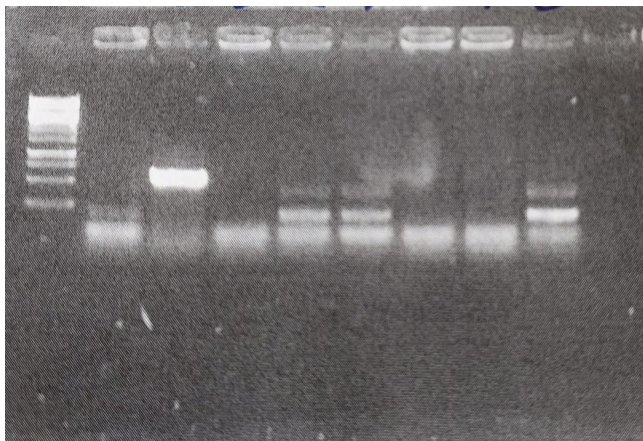
HFBII: 1= ladder, 2 to 8=colonies

HFBI.jpeg



HFBI: 1=ladder, 2 to 8= colonies

MBSP1.jpeg



MBSP1: 1=ladder, 2 to 8= colonies

negative result: around a 900 bp band was expected as a negative result.

positive result: HFBI= 1600 bp & HFBII=1714 bp & MBSP1= 2317 bp

result: as seen in the results of our gel electrophoresis we did not get the desired amplicon in many of our colonies and as we were short of time the few that could have been successfully transformed could not be sent for sequencing to be able to proceed with *S Cerevisiae* transformation.

participants: Junfei Chen and Shekoufeh M M Sadeghi